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Impaired autobiographical remembering and future thinking in temporal lobe epilepsy

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Abstract

Objective: The ability to imagine future events, episodic simulation (ES), is critical for goal setting and adaptive functioning. Although ES engages many of the same neural substrates as episodic memory (EM) in healthy controls, particularly the hippocampus, its disruption in clinical populations remains poorly characterized. Temporal lobe epilepsy (TLE) is a condition marked by mesial temporal pathology and known EM impairments. However, the relationship between EM and ES deficits and hippocampal structural correlates remain unexplored. Here, we tested whether TLE is associated with impoverished ES beyond EM and language impairments and evaluated the relationship between ES and hippocampal morphology.

Methods: We assessed patients with drug-resistant TLE and controls using the adapted autobiographical interview and hippocampal morphometry (subfield volume, thickness, and gyrification) derived from HippUnfold.

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Author Contributions

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Patient Consent Statement & Ethics Approval Statement

This study, informed consent, and all procedures were approved by the Institutional Review Board at the Medical University of South Carolina and in accordance with the Declaration of Helsinki.

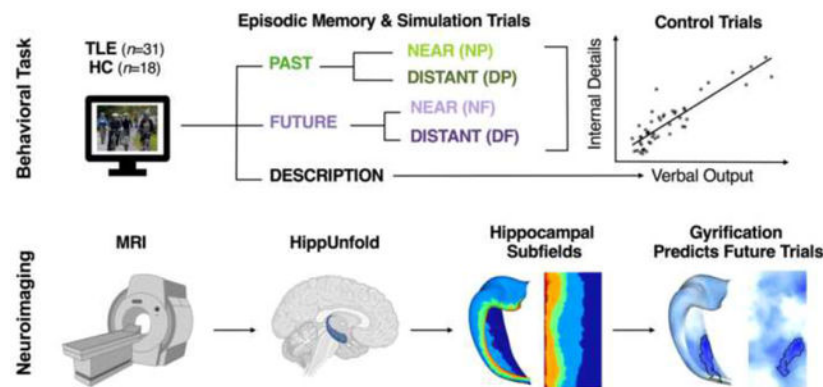
Declaration of Competing Interests

The authors report no conflicts to disclose.

Results: TLE patients exhibited marked deficits in internal detail generation for both past and future events, even after controlling for verbal output. Performance on episodic memory and simulation showed weak coupling in TLE. Left TLE patients showed uniformly reduced internal detail generation, while right TLE patients retained a past–future gradient. Subfield volumes (left CA1 and right CA2/3) predicted EM performance in right TLE only, whereas left hippocampal gyrification predicted performance across both EM and ES. The cluster predicting ES was more specific, as whole hippocampal gyrification predicted EM but not ES.

Conclusion: These findings reveal overlapping but differing hippocampal correlates for deficits in remembering and imagining episodic details in TLE and underscore the clinical relevance of assessing future-oriented cognition.

Graphical Abstract



Keywords

Temporal lobe epilepsy; episodic memory; hippocampal subfields; hippocampal morphometry

1 Introduction

Episodic memory (EM) [1] and episodic simulation (ES) refer to the capacity to recall and re-experience past events and imagine plausible future autobiographical scenarios, respectively [2]. These interrelated processes are essential for adaptive functioning [3], including planning, goal setting, decision-making, and emotion regulation [4]. While EM has been extensively studied in neurological populations, the clinical relevance of ES remains underexplored despite its centrality to everyday functioning.

Autobiographical episodic memory and simulation refer to personally experienced or imagined events, in contrast to laboratory-based episodic tasks such as word lists or story recall [5]. We will refer to autobiographical (i.e., personal) episodic memory/simulation as episodic memory/simulation from this point on, though we recognize episodic memory can be defined in several ways. Within this autobiographical framework, according to the constructive episodic simulation hypothesis, ES extracts and recombines elements from episodic memory into novel future scenarios. Thus, EM impairment constrains ES as fewer stored details are available for recombination [2].

Autobiographical deficits are well documented across a range of neurological conditions with hippocampal and/or PFC involvement, including traumatic brain injury [6], Alzheimer's disease [7], fronto-temporal dementia [8], and amnesia [9]. Temporal lobe epilepsy (TLE) [10] offers a natural lesion model for investigating the neural basis of EM and ES due to its hallmark temporal lobe pathology, most prominently, hippocampal atrophy. Further, TLE patients have a *gradient* of hippocampal damage, from subtle volume loss to hippocampal sclerosis, allowing for detection of brain-behavior relationships. While these processes and deficits overlap in healthy individuals [11–14] and aging [15], evidence from clinical populations demonstrates the processes may be dissociable. Prior work in patients with ventromedial PFC lesions has shown preserved memory for past events alongside impaired simulation of the future [16]. Observed dissociations involve primarily frontal pathology, and it is unknown if populations with hippocampal damage show similar or different patterns in EM and ES deficits.

Patients with TLE consistently exhibit deficits in autobiographical memory, particularly in generating rich, specific details [10, 17–19]. The influence of seizure laterality on EM performance remains debated: left TLE is often associated with greater impairments, particularly for remote memories [20], but some studies have reported greater deficits in right TLE [21]. Functional imaging in TLE patients undergoing anterior temporal lobectomy (ATL) reveals that, while healthy controls and right TLE patients often engage a predominantly left-lateralized autobiographical memory network, left TLE patients often recruit contralateral regions, possibly reflecting compensatory mechanisms [20]. These inconsistencies underscore unresolved questions about hemispheric specialization and the anatomy of autobiographical cognition [22].

TLE patients also experience deficits in ES [17, 23], including impoverished detail generation and reduced construction [24], particularly in left-sided cases [25]. These impairments have been partially attributed to difficulties with lexical retrieval [26], but several questions remain unanswered. First, few studies have systematically controlled for verbal output, making it difficult to distinguish genuine episodic impairments from reductions in language production, because reduced verbal output limits the extent of retrieved details. Next, theoretical models suggest that EM and ES rely on shared mechanisms, namely the retrieval of stored episodic details and their flexible recombination into coherent scenes [2, 7]. However, it is unclear whether this extends to clinical populations.

In addition to a behavioral relationship, EM and ES rely on overlapping neural networks [27] spanning frontal, temporal, and parietal regions, with the hippocampus and prefrontal cortex (PFC) as central hubs [11, 28, 29]. Functional neuroimaging studies in healthy individuals consistently implicate these structures in autobiographical cognition [11–14], and lesion studies confirm their critical role for producing both past and future content [30, 31]. Within the hippocampus, anterior and posterior segments may support distinct but complementary roles in healthy populations: posterior regions support the retrieval of episodic details while anterior regions integrate retrieved episodic details into coherent scenarios [32]. Preliminary evidence from healthy adults suggests hippocampal subfields may contribute differentially to episodic processing. A recent fMRI study reported that

CA1 representations generalized across items encountered within the same episode while CA2/3/DG representations differentiated them [33]. A separate volumetric study found posterior CA2/3 volume related to autobiographical recall ability among healthy adults with weaker recall [34]. However, the structural correlates of EM and ES impairment in TLE are poorly characterized, particularly for ES. Tools such as HippUnfold now enable surface-based morphometric analysis of the hippocampus, including thickness and gyrification, a measure of hippocampal folding complexity, which may capture subtle structural variation in TLE not reflected in whole-volume measures [35, 36]. Early work using these methods has demonstrated their sensitivity to hippocampal pathology and relevance to memory organization in TLE [37], but whether these features relate to autobiographical cognition has not been tested.

The current study addresses these gaps by evaluating EM and ES in patients with unilateral, drug-resistant TLE compared to healthy controls, while systematically controlling for verbal output. We tested the relationship between EM and ES deficits in TLE and whether specific hippocampal morphometric features predict performance in this poorly understood but clinically relevant cognitive domain (i.e., episodic simulation). We also examined whether disease laterality influences autobiographical performance. Our primary hypotheses concerned the behavioral relationship between EM and ES in TLE (H1, H2). We additionally tested laterality effects (H3) and structural correlates (H4) to contextualize these behavioral findings.

Primary hypotheses:

1. *H1*: Patients with unilateral TLE will demonstrate impaired episodic simulation beyond episodic memory deficits as compared to controls, even when controlling for verbal output.
2. *H2*: Episodic memory and episodic simulation performance will show limited behavioral coupling in TLE.

Exploratory hypotheses:

3. *H3*: Left TLE will be associated with more uniform impairment across episodic memory and simulation as compared to right TLE.
4. *H4*: Structural hippocampal correlates will differentially relate to episodic memory versus simulation performance in the TLE cohort.

Through this work, we aim to (1) systematically examine the relationship between EM and ES at the behavioral and neural level, (2) advance the understanding of how the hippocampus supports EM and ES in TLE, and (3) highlight the clinical relevance of assessing future-oriented thinking in TLE. Understanding whether future-oriented cognition is impaired independently of memory in TLE has direct implications for presurgical counseling, functional outcome prediction, and post-operative quality-of-life assessments.

2 Materials & Methods

Patients included were between 18 and 75 years of age and diagnosed with unilateral drug-resistant TLE based on the International League Against Epilepsy criteria [38, 39]. We included drug-resistant cases as these patients undergo thorough work-up with 3T MRI and prolonged video EEG monitoring to confirm the diagnosis and characterize the laterality and lobar origin of seizures. Thus, patients included demonstrated clinical features typical of TLE and had radiographic or neurophysiological confirmation, or both. Only patients whose primary language was English, had no history of developmental delay, and were able to provide informed consent were included. Healthy controls were recruited locally from a cohort of neurotypical adults (≥ 18 years old) with no history of neurological or psychiatric conditions. This study, informed consent, and all procedures were approved by the Institutional Review Board at the Medical University of South Carolina (MUSC) and in accordance with the Declaration of Helsinki.

2.1 Sample Demographics

Thirty-one patients with drug-resistant TLE ($n=17$ left TLE, $n=14$ right TLE) and 18 healthy controls participated in the study. Controls were on average 30.78 years old ($SD=9.05$), and patients were on average 39.48 years old ($SD=11.11$). This age difference was statistically significant, with patients being older than controls ($t=-2.98$, $p=.005$) and, thus, we used age as a covariate when comparing TLE vs. controls. Sex distribution did not differ significantly between groups ($\chi^2=0.39$, $p=0.53$): 83% of controls and 71% of patients were female.

2.2 Autobiographical Interview Administration & Processing

Participants completed the adapted autobiographical interview (AAI) [15] using picture cues [40], designed to elicit either recollection of a past (EM) event or simulation (ES) of a plausible future event (time direction). Examples of picture cues include photographs of the beach, cyclists in a park, or an ice-skating rink (Figure 1A). The AAI is a naturalistic test of autobiographical EM and ES [41], relying on self-generated episodic content in contrast to measures like story recall (i.e., logical memory). Events were cued at either near (weeks or months) or distant (years) timepoints (temporal distance). This yielded four experimental conditions: distant past (DP), near past (NP), near future (NF), and distant future (DF), each consisting of three trials with unique picture stimuli (Figure 1). Our control condition was a picture description task (6 trials) during which participants were instructed to describe a target picture in as much detail as possible for up to three minutes. Picture stimuli were randomly assigned to trials across participants, and each participant viewed each image only once to avoid repetition across conditions.

All transcripts were audio recorded and transcribed verbatim manually. An automated natural language processing algorithm quantified the number of internal and external details, which has been validated against gold standard manual scoring [42]. Internal details were defined as those that relate directly to the main event of the memory or simulation, such as details regarding time, place, perception, and emotions. External details were non-episodic and more generic in nature, including commentary, facts and references to events other than the target event, or semantic rather than episodic content. When internal details are

difficult to retrieve, individuals may compensate by relying more on semantic content [32, 41]. Following the protocol from the AAI automated scoring validation study [42], we manually scored 20% of the responses using a well-established standardized scoring manual developed in prior AI work [41]. Manual and automatic scoring showed excellent agreement for internal details ($r=0.85$, $p<.001$) and moderate agreement for external details ($r=0.57$, $p<.001$). Agreement for external details is typically lower than that for internal details; however, our primary analyses focused on internal details rather than external, as external details had low agreement and are not episodic in nature [43]. Thus, analyses on external details should be interpreted with caution. All analyses used internal details according to the automated scoring. A limitation of current automated scoring is that it only includes scoring of internal and external details, meaning specific types of details like perceptual or emotional are not yet available for analysis.

2.3 MRI Acquisition & Processing

Structural MRI scans were acquired on a 3T Skyra scanner (12-channel head coil) at MUSC. The MPRAGE T1-weighted protocol used the following parameters: TR=1900 ms, TE=2.36 ms, TI=900 ms, FOV=256 mm, flip angle=9°, voxel size=1×1×1 mm³. T1 weighted images were processed using HippUnfold [36], a pipeline that topologically unfolds the hippocampus and applies a U-net-based segmentation of the bilateral hippocampi into its subfields: CA1, CA2, CA3, CA4, dentate gyrus, and subiculum (Figure 2). HippUnfold provides vertex-wise measurement of thickness, gyrification, and curvature to capture detailed morphometric properties of the hippocampus. We generated HippUnfold outputs at 1mm resolution and applied a 2mm full-width at half-maximum (FWHM) Gaussian smoothing to reduce extreme surface values while preserving spatial resolution per previous work[37]. Although HippUnfold enables anatomically informed subfield estimation from standard-resolution T1-weighted images, subfield boundaries cannot be directly visualized at 1 mm³ resolution and results should be interpreted as morphometric proxies rather than histologically precise measures [44]. Visualizations of the segmented hippocampi were generated using HippoMaps [45]. Hippocampal abnormality, including mesial temporal sclerosis, was present in 9 of 17 left TLE and 8 of 14 right TLE patients (Table 2). Instead of stratifying analyses, we used continuous subfield volumes and vertex-wise gyrification, capturing more graded variation of the hippocampus rather than a binary classification. This approach is supported by recent work showing that binary clinical classification obscures the continuous gradient of underlying pathology in TLE [46].

2.4 Statistical Analyses

2.4.1 Behavioral: Behavioral and subfield volume analyses were performed in R Studio [47]. The primary dependent variable was the number of internal details produced during the AAI. Because autobiographical tasks inherently rely on verbal expression, a non-autobiographical verbal control condition is necessary to disentangle reductions in episodic detail from general differences in speech production. This is an underutilized but critical standard in the analysis of autobiographical recall and simulation [48], and its inclusion in this study addresses a major known limitation in prior analyses of autobiographical cognition in TLE. Verbal output was defined as the average number of words produced during the picture description trials. The picture description task represents an appropriate

control task because it has similar demands to the other trials but eliminates the presence of autobiographical content (for further discussion, see [26]). Verbal output significantly correlated with internal details ($\rho=0.83$, $p<.001$, 95% CI [0.72, 0.90]) and external details ($\rho=0.75$, $p<.001$, 95% CI [0.60, 0.85]). The strong correlation between picture description output and internal detail production supports the use of this measure as an approximation for general verbal capacity; however, we report both corrected and uncorrected results as verbal output could potentially obscure or reveal episodic features.

An initial ANCOVA examined group differences (TLE vs. HC) in the number of internal details produced across time direction (past vs. future) and temporal distance (near vs. distant) while controlling for verbal output and age. Next, we applied a similar ANCOVA to examine differences in number of internal details produced across laterality (left vs. right TLE), time direction, and temporal distance, while controlling for verbal output and duration of disease, a commonly used clinical covariate when evaluating memory and language in TLE [49]. We reran each ANCOVA omitting the verbal output covariate to demonstrate that covariate adjustment did not obscure true group differences through removal of shared episodic variance. We then used partial correlations to test whether average internal detail generation for past trials predicted average internal detail production for future trials, and vice versa, while controlling for verbal output. Partial correlations were run separately in patients and controls to assess the degree of behavioral coupling in each group.

2.4.2 Neuroimaging: All neuroimaging analyses were conducted in TLE patients and included laterality as a covariate, as contralateral hippocampal reorganization is documented in TLE [20, 37]. Imaging was available for 28 of the 31 patients. For volumetric neuroimaging analyses, we used R to examine the average number of internal details per condition as the dependent variable for past and future trials while controlling for verbal output and TLE laterality. For subfield analyses, we took a hypothesis driven approach examining left and right CA1 and CA2/3 volumes based on previous work linking these volumes to episodic and autobiographical memory performance [33, 34] (i.e., we did not examine CA4, subiculum, or dentate volume). We used linear models to relate the internal detail production to left and right CA1 and CA2/3 volumes, including an interaction term for volume $\times$ laterality while controlling for verbal output. We ran separate models for average internal details produced for past and future trials, correcting for multiple comparisons using false discovery rate (FDR) given our sample size. We excluded observations with a standard deviation greater than 3 from the mean for volumetric measures.

For our data driven vertex-wise (surface based) analyses, we used python to perform a vertex-wise general linear model (GLM) to assess hippocampal gyrification and thickness as well as dentate gyrus gyrification, using Freedman-Lane permutation-based clustering to control for multiple comparisons (5,000 permutations, p threshold= 0.01). Surface based analysis spanned the entire hippocampus: CA1, CA2, CA3, CA4, subiculum, dentate gyrus. These models identified regions where greater hippocampal thickness or gyrification predicted higher internal detail generation, controlling for verbal output and laterality. We then overlaid the significant clusters onto the HippUnfold subfield template to identify subfield location. In order to properly calculate effect sizes for neuroimaging analyses, we

examined the effect size of the whole hippocampal volume (sum of CA1, CA2, CA3, CA4, dentate gyrus, subiculum) relative to subfield volumes.

For vertex-wise analyses, we benchmarked cluster-level effects against the effect of whole-hippocampus mean gyrification using the same model structure. Specifically, for each FWE-corrected cluster, we computed each participant's mean gyrification across all vertices in the cluster and entered this cluster-mean as a predictor in a linear regression of internal details, controlling for laterality and verbal output. We computed partial R^2 for the cluster-mean effect, with 95% confidence intervals from a bias-corrected participant-level bootstrap (2,000 resamples). The whole-hippocampus benchmark used the same model with mean gyrification across all hippocampal vertices as the predictor. Because each analysis addressed distinct, prespecified hypotheses, corrections were applied within each family of tests (e.g., FDR for subfield analyses).

3 Results

3.1 Internal details & verbal output

Controls significantly outperformed TLE patients in both internal details and verbal output (Figure 1C–D). During picture description trials (control condition), controls produced an average of 220 ($SD=131$) words, compared to 77 ($SD=45$) words in patients, a measure we refer to as *verbal output* and use as a covariate. Across trials, controls generated more internal details ($M=148.8$, $SD=36.20$) than patients ($M=64$, $SD=11$).

3.2 Internal details across patients and controls (primary hypothesis 1)

An ANCOVA assessed internal detail production by group (TLE vs. HC), time direction (past vs. future), and temporal distance (near vs. distant) while controlling for verbal output and age. We also included interaction terms for group by time direction and group by temporal distance (Table 1). Patients produced significantly fewer internal details than controls ($p=0.01$, Cohen's $d=0.50$, 95% CI [0.11, 0.88]). There was a significant main effect of time direction as well as an interaction between group and time direction across past and future trials. There was not a main effect of temporal distance, but there was an interaction between group and temporal distance across near and distant trials. Verbal output was a significant covariate ($p<.001$, $\eta^2=0.63$, 95% CI [0.49, 1.00]). Age was not associated with production of internal details. These findings demonstrate reduced internal detail production in TLE, modulated by time direction and temporal distance (Figure 1). Patients also showed less variability than controls, demonstrating an established pattern in controls of producing more details for the past than future time points (Figure 1). An ANCOVA for external details by group can be found in Supplementary Table 1, which shows that external details did not differ across group (all $p>0.05$).

3.3 Weak behavioral coupling between episodic memory and simulation in TLE (primary hypothesis 2)

To examine the relationship between internal detail generation in past versus future trials we performed partial correlations separately for patients and controls controlling for verbal output. For TLE patients, performance on past trials did not significantly predict

performance on future trials ($R^2=0.10$, $p=0.09$, 95% CI [0.003, 0.37]). In controls, past performance accounted for a greater proportion of variance in future performance ($R^2 = 0.21$, $p=0.07$, 95% CI [0.001, 0.59]) than in TLE patients ($R^2 = 0.10$), suggesting relatively stronger coupling in the control group. Independent of significance testing, there is limited shared variance between past and future autobiographical performance in patients compared to controls (Figure 2A).

3.4 Internal details across laterality (exploratory hypothesis 3)

Seventeen patients (55%) had left-sided seizure onset and 14 (45%) had right-sided onset (Table 2). Age of onset and disease duration did not correlate with internal detail production ($r=-0.004$, $p=0.98$; $r=-0.15$, $p=0.43$).

A second ANCOVA examined internal detail production by seizure laterality (left vs. right TLE), time direction (past vs. future), and temporal distance (near vs. distant) while controlling for verbal output and duration of disease. We included interaction terms for laterality by time direction and laterality by temporal distance (Table 1). There was no main effect of laterality ($p=0.27$) on internal detail production. There was a main effect of time direction ($p<.001$) but not temporal distance ($p=0.15$). There was a significant interaction between laterality and time direction across past and future trials. There was a significant interaction between laterality and temporal distance across near and distant trials. Verbal output showed a main effect ($p<.001$; $\eta^2=0.49$, 95% CI [0.26, 1.00]). Disease duration was not associated with internal detail production ($p=0.64$). As shown in Table 1 and visualized in Figure 1, right TLE patients preserved both time direction and temporal distance gradients, producing more details for past than future events and for distant than near events, while left TLE patients showed uniformly reduced performance across all conditions (Figure 1E–F, Table 1). These interaction effects indicate differing modulation of autobiographical detail by time direction and distance across laterality (ANCOVA for external details in Supplementary Table 1). Critically, the ANCOVA that excluded verbal output no longer had an effect of time direction, demonstrating the importance of including a measure of verbosity in order to isolate true differences in episodic processing.

3.5 Hippocampal subfield volumes (exploratory hypothesis 4)

Next, we examined whether CA1 and CA2/3 volumes were associated with internal detail production for past and future trials for patients with neuroimaging data ($n=28$). There was one volumetric outlier in right CA1 greater than 3 standard deviations from the mean, which was removed from the analysis of right CA1. We used linear models with internal details as the dependent variable, verbal output as a covariate, and an interaction term for volume and laterality with separate models for past and future trials, and FDR correction was applied across the four subfield–hemisphere combinations. There was a significant interaction between left CA1 volume and laterality ($t=2.75$, $p=0.037$, partial $\eta^2=0.24$, 95% CI [0.02, 0.49]) as well as right CA2/3 volume and laterality ($t=2.54$, $p=0.037$, partial $\eta^2=0.21$, 95% CI [0.01, 0.47]) for past trials but not future trials ($p>0.05$). These interactions indicated that larger subfield volumes supported production of internal details for right TLE patients only (Figure 2C–D). Neither right nor left whole hippocampal volume predicted past or future trial performance in right or left TLE ($p>0.05$).

3.6 Hippocampal thickness & gyrification (exploratory hypothesis 4)

In our whole hippocampus data-driven approach, vertex-wise GLMs examined clusters of hippocampal thickness and gyrification that positively predicted internal detail production, controlling for verbal output and TLE laterality for patients with neuroimaging data ($n=28$). There were no significant clusters in the dentate gyrus and no significant clusters in thickness of the hippocampus. Significant clusters were only observed in left hippocampal gyrification (Figure 3). For past trials, increased gyrification in left CA1 (60% of the cluster) and the subiculum (40% of the cluster) along the mid to posterior hippocampus predicted internal detail production ($t = 3.09$, partial $R^2 = 0.28$, 95% CI [0.04, 0.59], 87 vertices). For future trials, increased gyrification in the left subiculum (84% of the cluster) and CA1 (16% of the cluster) along the mid to posterior hippocampus predicted internal detail production ($t = 3.32$, partial $R^2 = 0.32$, 95% CI [0.10, 0.52], 92 vertices). Whole hippocampal gyrification predicted past trial performance (partial $R^2 = 0.21$, $p = 0.02$), with the cluster effect (partial $R^2 = 0.28$) somewhat larger. For future trials, the cluster effect was substantially larger than the whole-hippocampus effect (cluster partial $R^2 = 0.32$ vs. whole HPC partial $R^2 = 0.12$, $p = 0.08$), suggesting the cluster captures spatially specific variation related to future thinking that is diluted when averaging across the whole hippocampus.

4 Discussion

The current study investigated both episodic memory and episodic simulation in patients with TLE, comparing their performance to that of healthy controls. We found that patients with TLE consistently generated fewer internal details on both EM and ES tasks, even when controlling for verbal output and age. Additionally, EM and ES showed weak behavioral coupling in TLE. These behavioral impairments reflect a flattened capacity to retrieve and simulate richly detailed experiences, which are core to planning, goal setting, and functional independence. Clinically, our findings underscore the relevance of assessing future-oriented cognition in TLE.

Our exploratory hypotheses revealed that left TLE patients exhibited more severe impairments than right TLE patients, who retained a normative gradient similar to controls. At the subfield level, we found that subfield volumes, specifically left CA1 and right CA2/3, predicted production of internal details, but only in right TLE patients. In contrast, vertex-wise models of gyrification identified consistent predictors of internal detail production in the left hippocampus across memory and simulation tasks. Hippocampal subfield volumes predicted internal detail generation for past trials only, and the gyrification clusters predicting EM and ES overlapped but were not identical. Overall, the specificity of the cluster for future trials was higher as whole hippocampal gyrification predicted past trials but not future. These findings broadly indicate that both regional volume and local hippocampal morphometric features contribute to autobiographical cognition in TLE, with structure-function relationships that differ by seizure laterality and past versus future trials.

4.1 TLE patients vs. controls (primary hypothesis 1)

Our findings replicate established evidence of EM impairments in TLE patients compared to controls [10, 17, 18] and extend these findings through systematically controlling for verbal

output when examining group differences, addressing a known limitation in the literature of autobiographical cognition [48]. Because patients with TLE consistently produced fewer words across all trial types, controlling for verbal output allowed us to isolate genuine episodic deficits from reductions in verbal fluency or individual differences in verbosity. Our findings are concordant with previous studies, such as Rayner et al. (2022) [26] and Lechowicz et al. (2016) [25], highlighting impoverished future thinking in TLE. Overall, these studies report reduced sensory-perceptual detail generation more markedly in left TLE patients, emphasizing scene construction deficits and identifying lexical access challenges during episodic simulation [28]. Our findings corroborate and expand on these results by showing flattened performance across past and future events (i.e., little variation seen across trial types that is observed in controls), with deficits observed regardless of time direction (past vs. future) or temporal distance (near vs. far). This lack of variation contrasts with controls, who typically produce richer detail for past versus future events and for near versus distant events [50]. Such “flattened” performance has also been observed in EM tasks [22], suggesting a general disruption of episodic processes in patients with TLE, an effect that remained after adjusting for age.

4.2 Weak behavioral coupling between episodic memory and simulation (primary hypothesis 2)

Importantly, we examined the relationship between EM and ES in TLE, as these processes behaviorally overlap in healthy controls; however, this relationship did not fully extend to TLE. Coupling of EM and ES was weaker in TLE than in controls ($R^2 = 0.10$ vs. 0.21), though neither reached significance. While we may be underpowered to detect small effects, the point estimate in patients ($R^2 = 0.10$) indicates limited shared variance even if significant in a larger sample. The difference in magnitude between patients and controls suggests that TLE may alter the typical relationship between remembering and imagining. According to the constructive episodic simulation hypothesis, simulation requires flexibly recombining elements from episodic memory into novel future scenarios, such that EM impairment should constrain ES [2]. However, dissociations between EM and ES have been documented following focal brain damage, particularly in patients with vmPFC lesions who show relatively preserved memory alongside impaired future thinking [16, 31]. This indicates that future-oriented cognition recruits multiple processes beyond episodic memory alone. Our findings extend this pattern to a clinical population with hippocampal pathology: TLE disrupts both processes, but the limited shared variance indicates that deficits in remembering do not fully explain variability in imagining within this sample. Larger samples will be needed to clarify the magnitude of this relationship.

4.3 Left vs. right TLE patients (exploratory hypothesis 3)

When analyzing disease laterality effects, we observed distinct patterns between left and right TLE patients. While both groups performed worse than controls, right TLE patients exhibited a typical pattern of producing more internal details for past versus future events and near versus distant time points even when controlling for verbal output and disease duration. In contrast, left TLE patients showed uniformly reduced internal detail generation across all conditions, significantly diverging from both controls and patients with right TLE. These findings are consistent with previous research showing greater EM impairments

in left TLE [10, 21, 22] and extend this vulnerability to ES. Interestingly, Rayner et al. reported dissociable mechanisms underlying ES deficits by TLE laterality, with scene reconstruction deficits in left TLE and broader semantic impairments in right TLE [26]. The current study shows that these findings persist even when controlling for verbal output. We interpret the interaction between laterality and time direction and temporal distance as modulation of autobiographical detail. Further, the main effect of time direction was only clear when controlling for verbal output, suggesting verbal output across trial types may obscure underlying episodic features. Raw internal detail counts therefore confound verbal output with episodic features, while the model controlling for verbal output isolates episodic-specific variance. Thus, controlling for verbal output is critical in TLE, particularly considering subtle verbal impairments are common, and previous reports have highlighted the confounding effect of language impairments on memory measures [51, 52].

In line with prior studies [22, 23], we found no significant relationship between internal detail generation and age at epilepsy onset or duration of disease. These findings suggest that episodic deficits in TLE are more closely related to laterality and extent of MTL pathology than the chronicity of disease. This observation emphasizes the need for further work to clarify how epilepsy-specific structural and functional changes impact autobiographical cognition.

4.4 Contribution of hippocampal volumes & gyrification (exploratory hypothesis 4)

From a clinical perspective, the laterality-specific structure–function relationships are particularly relevant, as they align with known asymmetries in autobiographical and semantic processing in TLE [21, 37, 53]. A notable contribution of this study is analysis of hippocampal subfields and morphometric features (i.e., thickness, gyrification) in relation to internal detail generation. Prior work has implicated CA1 and CA2/3 in episodic processing in healthy adults [33, 34], albeit through different paradigms; our findings broadly extend this to TLE, where left CA1 and right CA2/3 volumes predicted performance in right TLE patients. That subfield volumes supported autobiographical memory only in right TLE is not surprising, as this group was generally less impaired than left TLE patients, allowing for greater variability to detect brain-behavior relationships. Subfield volumes predicted internal detail generation for past but not future trials, suggesting that retrieving established memories and constructing novel simulations may draw on partially distinct neural mechanisms.

Beyond volume, our vertex-wise analysis revealed clusters of gyrification in the left hippocampus associated with internal detail production across TLE patients. Gyrification in left CA1 and the subiculum predicted past trials, and gyrification in mainly the subiculum predicted future trials, spanning the mid to posterior hippocampus. While gyrification for past and future trials does overlap, there are slightly different patterns of gyrification supporting behavioral performance. Further, our vertex-wise regression approach demonstrated that gyrification explained slightly more variance in EM and ES than subfield volumes. Clusters also revealed differing spatial specificity, as the cluster predicting future performance was more selective than the cluster predicting past trials; whole hippocampal gyrification predicted past trials at a similar level as the cluster. Given

the use of standard-resolution T1-weighted imaging, these subfield-specific effects should be interpreted as hypothesis generating regionally localized morphometric associations rather than precise cytoarchitectural dissociations.

Although prior task-based fMRI studies in healthy individuals often highlight anterior versus posterior hippocampal functional specialization [32, 54, 55], our structural approach found mid and posterior hippocampus gyrification to be associated with autobiographical abilities. Research examining hippocampal features such as thickness and gyrification remains limited due to the novelty of such methods. However, recent work using HippUnfold has demonstrated its utility in identifying hippocampal sclerosis and characterizing reorganization of episodic and semantic memory systems in TLE [35, 37]. For example, Cabalo and colleagues used a task-based fMRI memory paradigm and demonstrated that structural hippocampal alterations (mean diffusivity and thickness) were associated with poorer task performance [37]. While measurement of episodic memory differs from the current study and Cabalo and colleagues (who used an object pairing task), leveraging detailed analysis of the hippocampus shows clear advantages in both studies, revealing that disruptions in small portions of the hippocampus relate to behavior.

4.5 Implications

This study has potential implications for both research and clinical practice. From a research standpoint, TLE offers a natural model for studying the neural mechanisms of EM and ES due to its characteristic temporal lobe pathology. TLE patients have a *gradient* of hippocampal damage, from subtle volume loss to hippocampal sclerosis, which varies across patients and hemispheres, enabling detection of brain-behavior relationships. Our findings emphasize the need to consider hippocampal subfield and gyrification correlates of episodic detail generation. Impairments in episodic simulation may affect patients' ability to anticipate future scenarios, plan complex behaviors, and evaluate potential outcomes. In clinical contexts, this may influence adherence to medication regimens, engagement with treatment plans, and broader decision-making processes relevant to epilepsy management.

Clinically, episodic simulation and broader future-oriented thinking are rarely, if ever, assessed in neurological or neuropsychological settings, despite their critical role in adaptive functioning [4, 56]. Incorporating assessments of future thinking into clinical practice could enhance diagnostic evaluations and inform interventions aimed at improving quality of life for TLE patients. Impaired future thinking may potentially manifest as reduced ability to anticipate consequences, plan future actions, and adhere to complex treatment regimens such as medication schedules or post-surgical care. By demonstrating that EM and ES show weak behavioral coupling and may represent partially independent clinical impairments in TLE, we highlight the need to consider both processes when evaluating autobiographical cognition in clinical populations. However, the practical burden of scoring autobiographical tasks remains a barrier: administration requires 30–60 minutes, followed by full verbatim transcription and structured scoring. These constraints have limited widespread adoption. Our study demonstrates these hindrances can be alleviated through reliable, externally validated automated scoring. Paired with newer tools that allow for reliable word-by-word

transcriptions of oral speech, the field is now in an ideal position to explore these complex cognitive domains in the comprehensive assessment of patients' cognitive abilities.

4.6 Limitations & future directions

A number of limitations apply to this study. First, several limitations relate to our sample. Our modest sample size reflects the manual labor demands of the pre-automation era; the data were collected before the widespread availability of such reliable transcription tools, hence necessitating human word-by-word transcriptions. Future studies should aim to replicate these results in larger cohorts, taking advantage of newly available automatic transcription and scoring tools. Our control group was smaller than the patient group and was on average younger, and while age was included as a covariate in all group comparisons, this statistical adjustment cannot fully account for residual differences in cohort characteristics. Future work should incorporate age-matched controls. Language dominance was not available for all patients, limiting interpretation of laterality effects, particularly given the known interaction between language networks and autobiographical processing. Additionally, standardized neuropsychological measures were not available for all participants, preventing formal comparison between autobiographical performance and traditional memory metrics.

Second, several questions fall outside the scope of the present design. We recognize that because the picture description task shares some cognitive demands with autobiographical recall, it is possible that controlling for verbal output may remove variance that partially overlaps with episodic processing. Additionally, we did not explicitly assess real-world functional outcomes, limiting direct inference about the clinical impact of episodic simulation deficits. Next, the crosssectional design precludes conclusions about causality or changes over time. Longitudinal studies, particularly those incorporating test-retest over time as well as pre- and post-surgical evaluations, are needed to better understand the impact of epilepsy- and surgery-related effects on EM and ES.

Third, our analyses focused on specific measures and regions of the brain. Our study focused on internal details, more granular analyses of internal detail types (e.g., time, place, emotion) and semantic details (e.g., repetitions, generalizations) could yield deeper insights into the nature of autobiographical deficits in TLE and their neural bases. Similarly, future work should examine the association between EM and ES depending on temporal distance, separately examining recent and distant memories and simulations. We only examined features of the hippocampus, and despite finding varying predictors of EM and ES, analyses incorporating additional medial temporal as well as prefrontal structures could further elucidate structure-function relationships. Further, future research should also explore additional neuroimaging modalities and higher resolution (<1mm), such as diffusion-weighted imaging, to investigate structural connectivity disruptions underlying EM and ES deficits. Tools such as HippUnfold offer exciting avenues for analysis of white matter microstructure of the hippocampus with the added advantage of analyzing diffusion metrics in unfolded space. Combining these structural approaches with behavioral assessments and lesion-symptom mapping in longitudinal designs could provide a comprehensive understanding of the neural mechanisms underlying autobiographical

cognition in TLE. While the hippocampus is a critical region in EM and ES, whole brain analyses may reveal the neural bases of differing EM and ES performance across left and right TLE or altered connectivity between the hippocampus and EM and ES networks. Lastly, studies using fMRI in TLE could examine neural activation, as it is possible that the use of picture cues may impact results, as the posterior hippocampus is implicated more in visual processing and memory [57].

5 Conclusion

We found robust deficits in both episodic memory and episodic simulation in patients with TLE compared to healthy controls, even when controlling for age and verbal output. TLE patients also showed weaker coupling between episodic memory and episodic simulation compared to controls. Additionally, left TLE patients exhibited more severe impairments than right TLE patients. By linking autobiographical performance to hippocampal subfield volume and gyrification, our findings inform and extend clinical models of hippocampal dysfunction and autobiographical impairment in TLE, highlighting the clinical relevance of assessing future-oriented thinking in TLE.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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Data Availability Statement

All data and scripts used for analysis are available upon reasonable request from the corresponding author; however, restrictions may apply regarding patient consent and anonymity.

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Highlights

- Both episodic memory and simulation are impaired in TLE after controlling for verbal output
- Memory and simulation show weaker behavioral coupling in TLE than in controls
- Left TLE showed uniformly reduced performance, while right TLE preserved normative gradients
- Left CA1 and right CA2/3 volumes predicted memory performance in right TLE
- Left hippocampal gyrification predicted memory and simulation across TLE patients

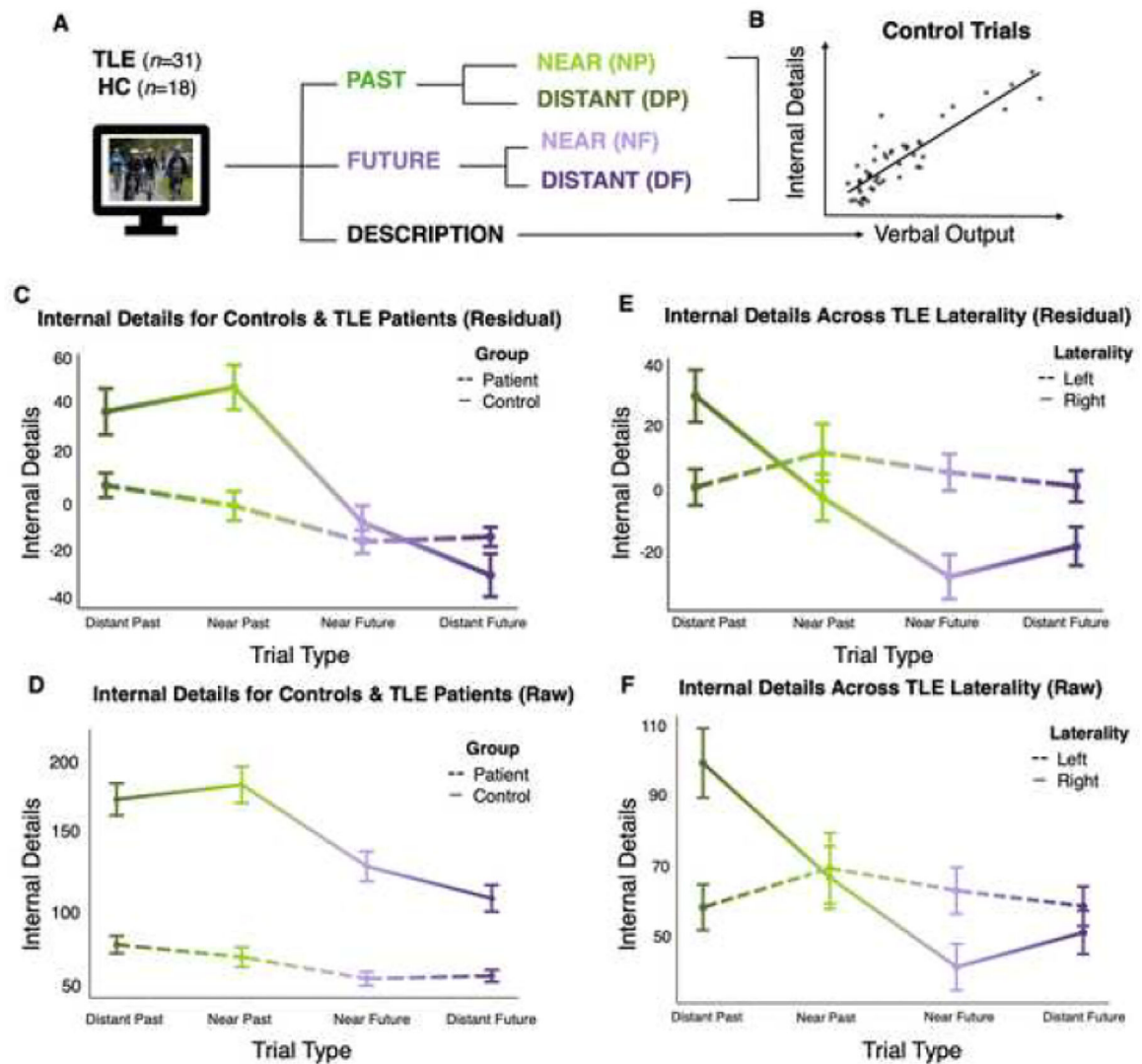


Figure 1. Episodic detail generation during memory and simulation tasks in TLE patients and controls.

(A). Task design: Participants (TLE=31; HC=18) viewed picture cues and generated descriptions for past and future events at near (NP, NF) and distant (DP, DF) time points, as well as a non-autobiographical picture description condition. (B). Total number of internal details correlated strongly with overall verbal output (C-D) TLE patients compared to controls controlling for verbal output (C, residuals) and raw internal details (D). TLE patients generated fewer internal details while (E-F) Left TLE patients showed flattened performance relative to right TLE (E) in raw internal details and when controlling for verbal output.

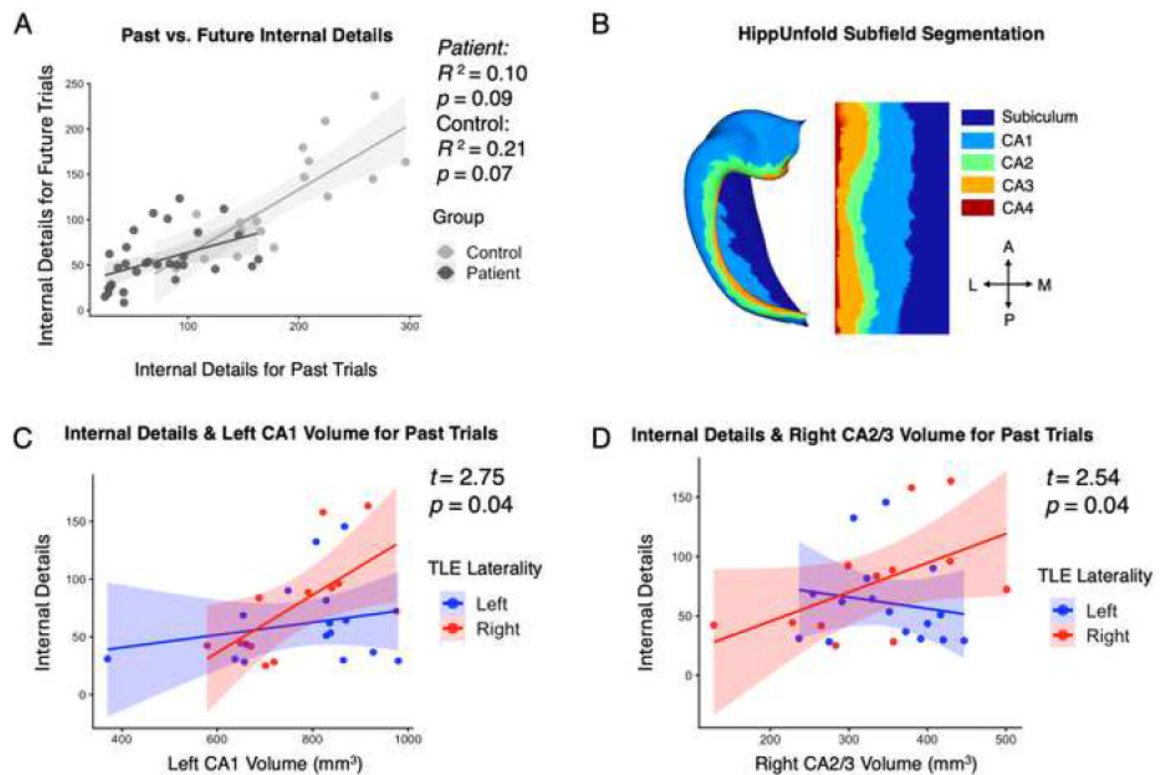


Figure 2. Weak behavioral coupling between episodic memory and simulation, example HippUnfold segmentation, subfield volumetric predictors of episodic detail generation in TLE patients.

(A) Internal details for EM (past) and ES (future) were weakly coupled in TLE patients ($p > 0.05$), demonstrating a weaker relationship between trial types than controls. Average internal details for past were predictive of future trials at a low level when examining both patients and controls while controlling for verbal output (residuals plotted). (B) Example image from HippUnfold showing unfolded hippocampal surface and subfield parcellation used for volumetric and morphometric analyses (Subiculum, CA1–CA4). (C–D) For past trials, left CA1 and right CA2/3 predicted internal detail production for past trials in right but not left TLE patients (not significant for left TLE or future trials, residuals plotted).

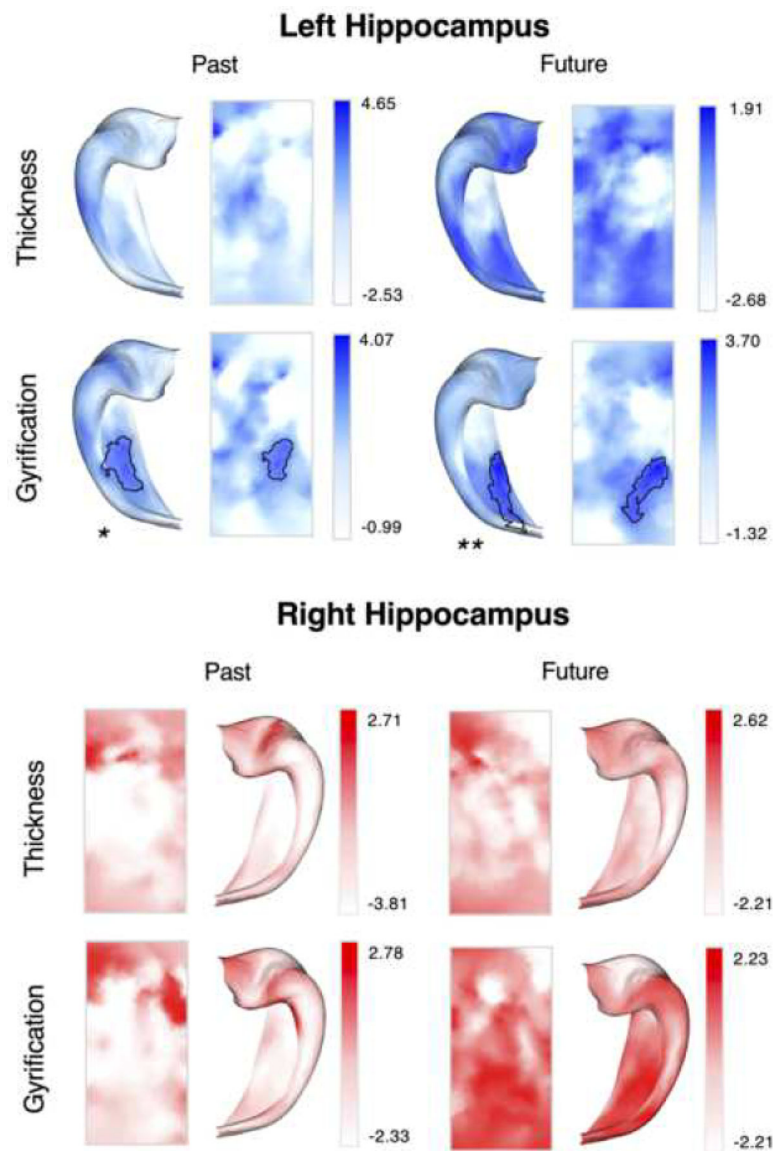


Figure 3. Left hippocampal morphometric features predict autobiographical detail generation in TLE.

Statistical maps showing vertex-wise associations between internal detail production and hippocampal gyrification across past * and future ** trials in TLE patients for the left and right hippocampus. Each panel shows 3D unfolded hippocampal surfaces and flattened 2D representations (right). Color bars reflect t-values; outlines indicate clusters surviving permutation-based correction (cluster extent thresholding). Significant predictors localized primarily to the subiculum and CA1.

Table 1.

ANCOVA for internal details in patients vs. controls and TLE laterality

Patient vs. Control Model		Including Verbal Output				Excluding Verbal Output			
		Sum of Squares	Degrees of Freedom	F value	p-value	Sum of Squares	Degrees of Freedom	F value	p-value
Age Group Verbal Output Temporal Distance Time Direction Group * Temporal Distance Group * Time Direction	Age	3395	1, 45	1.41	0.24	3572	1, 46	1.49	0.23
	Group	16407	1, 45	6.83	0.012	83929	1, 46	34.95	<.001
	Verbal Output	186668	1, 45	77.74	<.001				
	Temporal Distance	3804	1, 535	1.58	0.21	3804	1, 535	1.58	0.21
	Time Direction	210755	1, 535	87.77	<.001	210755	1, 535	87.77	<.001
	Group * Temporal Distance	14845	1, 535	6.18	0.013	14845	1, 535	6.18	0.013
	Group * Time Direction	63177	1, 535	26.31	<.001	63177	1, 535	26.31	<.001
TLE Laterality Model									
		Including Verbal Output				Excluding Verbal Output			
		Sum of Squares	Degrees of Freedom	F-value	p-value	Sum of Squares	Degrees of Freedom	F value	p-value
Duration Laterality Verbal Output Time Direction Temporal Distance Laterality * Time Direction Laterality * Temporal Distance	Duration	413	1, 27	0.23	0.64	6.8	1, 28	0.004	0.95
	Laterality	2279	1, 27	1.24	0.27	76.8	1, 28	0.042	0.84
	Verbal Output	48322	1, 27	26.36	<.001				
	Time Direction	34527	1, 337	18.81	<.001	441.2	1, 337	0.24	0.62
	Temporal Distance	3904	1, 337	2.13	0.15	2929.1	1, 337	1.60	0.21
	Laterality * Time Direction	24833	1, 337	13.53	<.001	24832.9	1, 337	13.53	<.001
	Laterality * Temporal Distance	18286	1, 337	9.96	0.002	18285.8	1, 337	9.96	<.001

Table 2:

Clinical variables for left and right TLE patients

	Left TLE	Right TLE
	Mean (SD)	Mean (SD)
<i>Age</i>	38.47 (10.81)	40.71 (11.7)
<i>Age at Onset</i>	26.68 (12.52)	25.5 (17.26)
<i>Duration</i>	11.79 (12.23)	15.21 (14.4)
<i>Seizure Frequency *</i>	6.51 (14.98)	14.7 (43)
<i>Number of AEDs</i>	1.94 (0.66)	2.0 (0.76)
<i>Hippocampal Abnormality Count (Total)</i>	9 (17)	8 (14)

* Seizure frequency reflects a composite clinical estimate of partial and partial to secondary generalized seizures and thus should be interpreted with caution.